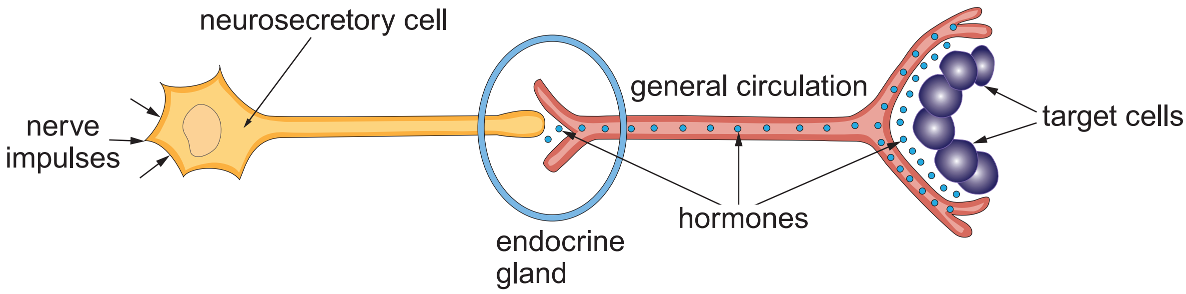
Examiner only

4. Osmoregulation is carried out by the kidneys, under hormonal control.
The regulatory process is represented in **Image 4.1**.

Image 4.1



(a) Using your knowledge of homeostasis and the mammalian kidney, identify each of the following shown in **Image 4.1**: [4]

(i) the source of the nerve impulses;

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(ii) the endocrine gland;

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(iii) the hormone;

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(iv) the precise location of the target cells.

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(b) Explain how the hormone involved in the process shown in **Image 4.1** affects the target cells and how it prevents excessive water loss. [3]

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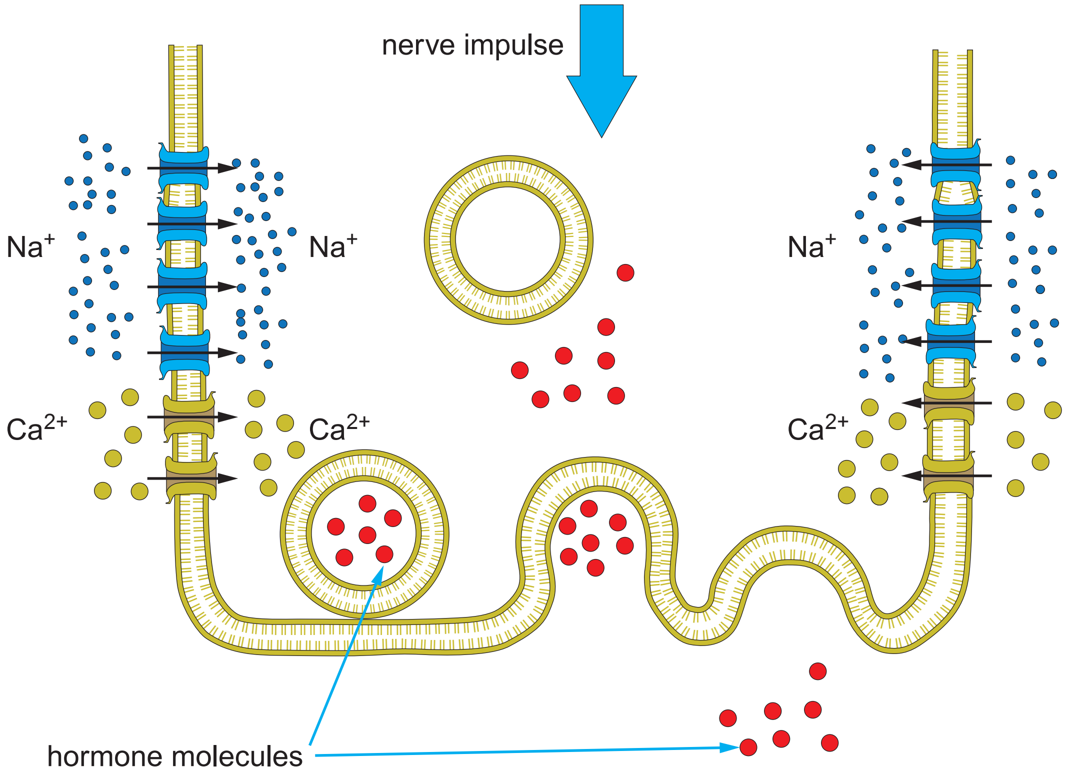
The release of the hormone is brought about by the arrival of a nerve impulse. This is similar to the way in which neurotransmitters are released. The process is shown in **Image 4.2**.

Image 4.2



(c) Using all the relevant information from both **Images 4.1** and **4.2** describe how the hormone molecules are released from the end of the neurosecretory cell. [5]

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Haemodialysis is one method of treating kidney failure.

Between dialysis sessions patients retain fluid and gain weight. Excess fluid is removed during the next dialysis session to return the patient to their target weight.

(d) Explain why it is important to remove excess fluid during haemodialysis. [2]

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